Part 1: The Ironies

- First, get into groups of **4 people (some groups will have 5).**
    - Introduce yourselves, if you don't know each other already.
    - As a group: decide on a specific application or domain where AI might be integrated (e.g. automated diagnosis, self-driving , LLMs/ChatGPT for daily use, etc). You will be using this throughout the exercise.
    - Assign each person in the group either A,B,C or D. You must have at least one of each. That person will read their assigned section. (If a group of 5, two people will read the same section).
    - First, everyone reads the introduction.
    - Then, read your sections individually and summarize it in your own words. Write down this summary. Then, when everyone is finished reading their sections, come back and share-around a summary of the section to your group-mates.
        - Then, discuss where you have encountered this irony yourself when you've worked with an automated/AI system.
        - Finally, discuss how this irony could arise in the application area you chose.
    - Finally, we'll all share our applications and the way these ironies can come up. Choose a representative to talk for your group.
    - **Feel free to mark up or add comments to the document to help your understanding.**
    - **When done: Group leader submit a PDF of the document to Gradescope. Make sure to export ALL TABS!**

- Some Vocabulary Notes
    - **VDU (Virtual Display Unit)** - This just means that the information is displayed on a computer screen (compared to physical dials and gauges). This paper was from the 1980s where this was a novel!

Group <NAME YOUR GROUP>

- Group Member Names
    - Member A : Frank
- **AI Application or Problem Chosen:**
    - Describe Problem Here

	Manual Control (A)	Cognitive Skills (B)	Monitoring (C)	Operator Attitudes (D)
Individual: Summarize this particular irony in	The fact that a machine made the operator less skilled but when it needs	First, the irony is that operators need time and	Human operators does not have the	A more complex system would require someone with more working

your own words. What is Bainbridge trying to say here?	fixing it needs a more skilled operator is ironic because a machine should ideally require less skills to operate but in the end more skill is needed.	experience to develop long-term knowledge but machines are taking that away from the person. The more reliant the person is on the machine, the less he is able to learn and troubleshoot when it breaks. Second, to fix the machine the person needs to have a working knowledge of it including knowing the current state of the machine. However, when a machine takes over it also takes over the ability for operators to develop a working knowledge thus making the fix harder.	attention span long enough to monitor if the machine is making the correct decision every second so sometimes an automatic control system is put in place because it does a better job than humans would. The irony here is that a human is also placed to check the control system which supposedly does a better job. Also, in the case where the machine makes a different decision than the human, which decision does it follow.	knowledge of the system to monitor and take over if needed. However, the irony here is that if the operator's job is only to monitor then he gets devalued and so is his pay because a job is based on the content rather than the skill.
When using any sort of automated system, have you personally encountered this irony before?	Yes. There are smart home appliances out there that are not as reliable as the non-smart ones and once they have a problem it is more expensive to fix them because you need someone more skilled and it is often more complicated to fix as well.	I think this is the case with modern AI systems. AI can write code/essays better and faster but if you rely on AI too much, you lose the ability to do so without it. So when you actually need to use the skill, it takes longer and more time because you haven't had the	In Boeing and AirBus aircrafts, they took two different approaches with the pilot. The latter is allowing the pilot to override any decisions that the machine made and the	I think similar things are happening for AI testers because to determine if the output code or knowledge is correct you need someone who is skilled in that area. But not a lot of people are willing to only monitor if they have specialized knowledge in that area because their job is devalued.

		ability to develop your working knowledge.	former is only taking in input from the machine.	
How could this irony arise in your particular application and lead to negative outcomes?	This process of making appliances smart is useless in most cases and most of the time only makes the process more complicated and takes longer to learn.	I think the overreliance on AI in this case is especially detrimental for programmers because once you rely on it you are not learning on your own and not improving your own coding skills.	This led to serious accident with Boeing 747 max because there is a bug with the control system of the plane that makes all the decisions but the pilot is unaware and it did not get noticed until after the plane crashed.	This could potentially lead to more errors in output for AI if they are not fact-checked. People can also be misled by AI systems if the outputs are incorrect.

Ironies of Automation

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Abstract

Irony : combination of circumstances, the result of which is the direct opposite of what might be expected.

Paradox : seemingly absurd though perhaps really well-founded statement.

The classic aim of automation is to replace human manual control, planning and problem solving by automatic devices and computers. However, as Bibby and colleagues (1975) point out : "even highly automated systems such as electric power networks, need human beings for supervision, adjustment, maintenance, expansion and improvement. Therefore one can draw the paradoxical conclusion that automated systems still are man-machine systems, for which both technical and human factors are important." This paper suggests that the increased interest in human factors among engineers reflects the irony that the more advanced a control system is, so the more crucial may be the contribution of the human operator.

This paper is particularly concerned with control in process industries, although examples will also be drawn from flight-deck automation. In process plants the different modes of operation may be automated to different extents, for example normal operation and shut-down may be automatic while start-up and abnormal conditions are manual. The problems of the use of automatic or manual control are a function of the predictability of process behaviour, whatever the mode of operation. The first two sections of this paper discuss automatic on-line control where a human operator is expected to takeover in abnormal conditions, the last section introduces some aspects of human-computer collaboration in on-line control.

1 Introduction

The important ironies of the classic approach to automation lie in the expectations of the system designers, and in the nature of the tasks left for the human operators to carry out.

The designer's view of the human operator may be that the operator is unreliable and inefficient, so should be eliminated from the system. There are two ironies of this attitude. One is that designer errors can be a major source of operating problems. Unfortunately people who have collected data on this are reluctant to publish them, as the actual figures are difficult to interpret. (Some types of error may be reported more readily than others, and there may be disagreement about their origin.) The second irony is that the designer who tries to eliminate the operator still leaves the operator to do the tasks which the designer cannot think how to automate. It is this approach which causes the problems to be discussed here, as it means that the operator can be left with an arbitrary collection of tasks, and little thought may have been given to providing support for them.

1.1. Tasks after automation.

There are two general categories of task left for an operator in an automated system. They may be expected to monitor that the automatic system is operating correctly and, if it is not, they may be expected to call a more experienced operator or to takeover themselves. We will discuss the ironies of manual takeover first, as the points made also have implications for monitoring. To take over and stabilise the process requires manual control skills, to diagnose the fault as a basis for shut down or recovery requires cognitive skills.

1.1.1. Manual control skills.

Several studies (Edwards and Lees, 1974) have shown the difference between inexperienced and experienced process operators making a step change. The experienced operator makes the minimum number of actions, and the process output moves smoothly and quickly to the new level, while with an inexperienced operator it oscillates around the target value. Unfortunately, physical skills deteriorate when they are not used, particularly the refinements of gain and timing. This means that a formerly experienced operator who has been monitoring an automated process may now be an inexperienced one. If they take over they may set the process into oscillation. They may have to wait for feedback, rather than controlling by open-loop, and it will be difficult for them to interpret whether the feedback shows that there is something wrong with the system or more simply that they have misjudged their control action. They will need to make actions to counteract their ineffective control, which will add to their work load. When manual takeover is needed there is likely to be something wrong with the process, so that unusual actions will be needed to control it, and one can argue that the operator needs to be more rather than less skilled, and less rather than more loaded [with task work and mental work], than average.

1.1.2. Cognitive skills.

Long-term knowledge. An operator who finds out how to control the plant for themselves, without explicit training, uses a set of propositions about possible process behaviour, from which they generate strategies to try (e.g. Bainbridge. 1981). Similarly an operator will only be able to generate successful new strategies for unusual situations if they have an adequate knowledge of the process. There are two problems with this for machine-minding operators. One is that efficient retrieval of knowledge from long-term memory depends on frequency of use (consider any subject which you passed an examination in at school and have not thought about since). The other is that this type of knowledge develops only through use and feedback about its effectiveness. People given this knowledge in theoretical classroom instruction without appropriate practical exercises will probably not understand much of it, as it will not be within a framework which makes it meaningful, and they will not remember much of it as it will not be associated with retrieval strategies which are integrated with the rest of the task. There is some concern that the present generation of automated systems, which are monitored by former manual operators, are riding on their skills, which later generations of operators cannot be expected to have.

Working storage. The other important aspect of cognitive skills in on-line decision making is that decisions are made within the context of the operator's knowledge of the current state of the process. This is a more complex form of running memory than the notion of a limited capacity short-term store used for items such as telephone numbers. The operator has in their head (Bainbridge, 1975) not raw data about the process state, but results of making predictions and decisions about the process which will be useful in future situations, including their future actions. This information takes time to build up. Manual operators may come into the control room quarter to half an hour before they are due to take over control, so they can get this feel for what the process is doing. The implication of this for manual takeover from automatically controlled plant is that the operator who has to do something quickly can only do so on the basis of minimum information. They will not be able to make decisions based on wide knowledge of the plant state until they have had time to check and think about it.

1.1.3. Monitoring.

It may seem that the operator who is expected solely to monitor that the automatics are acting correctly, and to call the supervisor if they are not, has a relatively simple task which does not raise the above complexities. One complexity which it does raise of course is that the supervisor too will not be able to take over if they have not been reviewing their relevant knowledge, or practising a crucial manual skill. Another problem arises when one asks whether monitoring can be done by an unskilled operator.

We know from many 'vigilance' studies (Mackworth. 1950) that it is impossible for even a highly motivated human being to maintain effective visual attention towards a source of information on which very little happens, for more than about half an hour. This means that it is humanly impossible to carry out the basic function of monitoring for unlikely abnormalities, which therefore has to be done by an automatic alarm system connected to sound signals. (Manual operators will notice abnormal behaviour of variables which they look at as part of their control task, but may be equally poor at noticing changes on others.) This raises the question of who notices when the alarm system is not working properly. Again, the operator will not monitor the automatics effectively if those have been operating acceptably for a long period. A classic method of enforcing operator attention to a steady-state system is to require him to make a log. Unfortunately people can write down numbers without noticing what they are.

A more serious irony is that the automatic control system has been put in because it can do the job better than the operator, but yet the operator is being asked to monitor that it is working effectively. There are two types of problems with this. In complex modes of operation the monitor needs to know what the correct behaviour of the process should be, for example in batch processes where the variables have to follow a particular trajectory in time. Such knowledge requires either special training or special displays.

The second problem is that if the decisions can be fully specified then a computer can make them more quickly, taking into account more dimensions and using more accurately specified criteria than a human operator can. There is therefore no way in which the human operator can check in real-time that the computer is following its rules correctly. One can therefore only expect the operator to monitor the computer's decisions at some meta-level, to decide whether the computer's decisions are 'acceptable'. If the computer is being used to make the decisions because human judgement and intuitive reasoning are not adequate in this context, then which of the decisions is to be accepted? The human monitor has been given an impossible task.

1.2. Operator attitudes.

I know of one automated plant where the management had to be present during the night shift, or the operators switched the process to 'manual'. This raises general issues about the importance of skill to the individual. One result of skill is that an operator knows they can takeover adequately if required. Otherwise the job is one of the worst types, it is very boring but very responsible, yet there is no opportunity to acquire or maintain the qualities required to handle the responsibilities. The level of skill that a worker has is also a major aspect of their status, both within and outside the working community. If the job is 'deskilled' by being reduced to monitoring, this is difficult for the individuals involved to come to terms with. It also leads to the ironies of incongruous pay differentials when the deskilled workers insist on a high pay level as the remaining symbol of status which is no longer justified by the job content.

Ekkers and colleagues (1971) have published a preliminary study of the correlations between control system characteristics and the operators' subjective health and feeling of achievement. To greatly simplify : high coherence of process information, high process complexity and high process controllability (whether manual or by adequate automatics) were all associated with low levels of stress and workload and good health, while fast process dynamics and a high frequency of actions which cannot be made directly on the interface were associated with high stress, high workload and poor health. High process controllability, good interface ergonomics and a rich pattern of activities were all associated with high feeling of achievement. Many studies show that high levels of stress lead to errors, while poor health and low job satisfaction lead to the high indirect costs of absenteeism, etc. (e.g. Mobley and colleagues, 1979).

Part 2: Approaches

- Individually, everyone read the Approaches section.
- Then everyone individually reads the passage corresponding with their letter. (Person E, choose one to make a trio)
- After you and your corresponding partner finish reading your section, summarize the points made.
- After you and your partner finish summarizing - share with the other half of your group.
- Think of your application or program that you mentioned in the previous question.
    - How could you incorporate some of Bainbridge's suggestions into it?
    - What effects do you think it would have?
    - Could the suggestion itself have a negative impact?
- Then choose a representative to share your application questions to the class.

	Knowledge Solutions (A & B)	Monitoring Solutions (C&D)
Pair: Summarize Bainbridge's points and suggestions in your own words.	Bainbridge argues that automation can leave human operators less prepared to intervene when failures occur. Operators who are not actively controlling a system may lack awareness of its current state and need time to understand what is happening before taking over. Very fast failures therefore require reliable automatic responses, while slower failures may allow intervention by well-practised operators. Automation also reduces opportunities to maintain manual and diagnostic skills, making regular practice and appropriate simulator training essential. Training must develop flexible problem-solving rather than just instruction-following, since procedures cannot anticipate every failure. The central irony is that the most reliable automated systems may	Bainbridge argues that humans cannot reliably monitor automation for rare failures without support, but that support must be carefully designed. Alarms should direct attention to important problems without overwhelming operators, and displays should provide clear information, with essential information still available if the computer fails. Automation should also reveal unusual changes and make failures obvious, since automatic adjustments can hide developing problems until they become serious. If operators are expected to check the computer's decisions, its reasoning and pace must be understandable, even at the expense of technical efficiency. Artificially introducing failures to keep operators alert can undermine trust and increase workload. Her central point is that assigning humans to supervise automation is only effective if the system supports their attention, understanding, and ability to intervene.

	require the greatest investment in operator training because they provide the fewest opportunities for humans to practise.	
Think of your application or program that you mentioned in the previous question. How could you incorporate some of Bainbridge's suggestions into it? What effects do you think it would have? Could the suggestion itself have a negative impact?	In an aircraft's autopilot system, Bainbridge's suggestions could be incorporated through regular manual flying practice, realistic simulator training, and clear displays that help pilots understand the aircraft's current state before taking over. Simulator training should include unexpected failures that require judgment rather than simply following memorized procedures. These measures could help pilots maintain their skills, recognize problems, and respond effectively when automation fails. However, manual practice during flight could increase workload and introduce opportunities for error, so it should occur under appropriate conditions. Training could also create false confidence if pilots assume that real emergencies will always resemble simulated scenarios.	In an aircraft's autopilot system, Bainbridge's suggestions could be incorporated through prioritized alarms, clear displays of what the autopilot is doing and why, and independent backup instruments that remain available if the main computer fails. The system could also alert pilots to unusual patterns of automatic corrections that might hide a developing problem. These features could help pilots notice failures sooner, understand the situation, and take over more effectively. However, excessive alerts could distract pilots or cause them to ignore warnings, while too much detail about automated decisions could increase their workload. The design should therefore make important changes clear without overwhelming pilots with information.

2. Approaches to solutions

One might state these problems as a paradox that by automating the process the human operator is given a task which is only possible for someone who is in on-line control. This section will discuss some possible solutions to problems of maintaining the efficiency and skills of the operator if they are expected to monitor and take over control : the next section will introduce recent proposals for keeping the human operator on-line with computer support.

Solving these problems involves very multidimensional decision making: suggestions for discussion will be made here. The recommendations in any particular case will depend on such factors as process size and complexity, the rate of process change, the speed and frequency of process or automatic control failure, the variability of the product and the environment, the simplicity and cost of shut down, and the qualities of the operator.

2.2. Working storage.

If the human operator is not involved in on-line control they will not have detailed knowledge of the current state of the system. One can ask what limitations this places on the possibility for effective manual takeover, whether for stabilization or shutdown of the process, or for fault diagnosis.

The straightforward solution when shutdown is simple and low-cost is to shut down automatically. The problems arise with processes which, because of complexity, cost or other factors (e.g. an aircraft in the air) must be stabilized rather than shutdown. Should this be done manually or automatically? Manual shut down is usable if the process dynamics can be left for several minutes while the operator works out what is happening. For very fast failures, within a few seconds (e.g. pressurized water nuclear reactor rather than an aircraft), when there is no warning from prior changes so that on-line working storage would also be useless, then reliable automatic response is necessary, whatever the investment needed, and if this is not possible then the process should not be built if the costs of failure are unacceptable.

With less fast failures it may be possible to 'buy time' with overlearned manual responses. This requires frequent practice on a high fidelity simulator, and a sufficient understanding of system failures to be sure that all categories of failure are covered. If response to failure requires a larger number of separate actions than can be made in the time available then some must be made automatically and the remainder by a highly practised operator.

2.3. Long-term knowledge.

Points in the previous section make it clear that it can be important to maintain manual skills. One possibility is to allow the operator to use hands-on control for a short period in each shift. If this suggestion is laughable then simulator practice must be provided. A simulator adequate to teach the basic behaviour of the process can be very primitive. Accurate fast reactions can only be learned on a high fidelity simulator, so if such reactions are necessary then this is a necessary cost.

Similar points can be made about the cognitive skills of scheduling and diagnosis. Simple pictorial representations are adequate for training some types of fault detection (Duncan and Shepherd, 1975), but only if faults can be identified from the steady-state appearance of the control panel and waiting for the steady-state is acceptable. If fault

detection involves identifying changes over time then dynamic simulators are needed for training (Marshall and Shepherd, 1981). Simple recognition training is also not sufficient to develop skills for dealing with unknown faults or for choosing corrective actions (Duncan, 1981).

There are problems with the use of any simulator to train for extreme situations. Unknown faults cannot be simulated, and system behaviour may not be known for faults which can be predicted but have not been experienced. This means that training must be concerned with general strategies rather than specific responses; for example simulations can be used to give experience with low probability events, which may be known to the trainer but not to the trainee. No one can be taught about unknown properties of the system, but they can be taught to practise solving problems within the known information. It is inadequate to expect the operator to react to unfamiliar events solely by consulting operating procedures. These cannot cover all the possibilities, so the operator is expected to monitor them and fill in the gaps. However, it is ironic to train operators in following instructions and then put them in the system to provide intelligence.
Of course, if there are frequent alarms throughout the day then the operator will have a large amount of experience of controlling and thinking about the process as part of his normal work. Perhaps the final irony is that it is the most successful automated systems, with rare need for manual intervention, which may need the greatest investment in human operator training.

2.1. Monitoring.

In any situation where a low probability event must be noticed quickly then the operator must be given artificial assistance, if necessary even alarms on alarms. In a process with a large number of loops there is no way in which the human operator can get quickly to the correct part of the plant without alarms, preferably also some form of alarm analysis. Unfortunately a proliferation of flashing red lights will confuse rather than help. There are major problems and ironies in the design of large alarm systems for the human operator (Rasmussen and Rouse, 1981).

Displays can help the operator to monitor automatic control performance, by showing the target values. This is simple for single tolerance bands, but becomes more complex if tolerances change throughout batch processing. One possible solution is to show the currently appropriate tolerances on a VDU by software generation. This does not actually get round the problems, but only raises the same ones in a different form. The operator will not watch the VDU if there is a very low probability of the computer control failing. If the computer can generate the required values then it should also be able to do the monitoring and alarms. And how does the operator monitor that the computer is working correctly, or take over if it obviously is not? Major problems may be raised for an operator who is highly practised at using computer generated displays if these are no longer available in an emergency. One ironic but sensible suggestion is that direct wired displays should be used for the main process information, and software displays for quantitative detail (Jervis and Pope, 1977).

'Catastrophic' breaks to failure are relatively easy to identify. Unfortunately automatic control can 'camouflage' system failure by controlling against the variable changes, so that trends do not become apparent until they are beyond control. This implies that the automatics should also monitor unusual variable movement. 'Graceful degradation' of performance is quoted in 'Fitts Lists' of man-computer qualities as an advantage of man over machine. This is not an aspect of human performance to be aimed for in computers, as it can raise problems with monitoring for failure (e.g. Wiener and Curry. 1980) : automatic systems should fail obviously.

If the human operator must monitor the details of computer decision making then, ironically, it is necessary for the computer to make these decisions using methods and criteria, and at a rate, which the operator can follow, even when this may not be the most efficient method technically. If this is not done then when the operator does not believe or agree with the computer they will be unable to trace back through the system's decision sequence to see how far they do agree.

One method of overcoming vigilance problems which is frequently suggested is to increase the signal rate artificially. It would be a mistake, however, to increase artificially the rate of computer failure as the operator will then not trust the system. Ephrath (1980) has reported a study in which system performance was worse with computer aiding, because the operator made the decisions anyway, and checking the computer added to his workload.

Part 3: Human Computer Collaboration

- Individually, everyone read the Human Computer Collaboration section.
- Then everyone individually reads the passage corresponding with their letter. (Person E, choose one to double up with)
- Summarize the points made, write down the summary, and then share with your group.
- Think of your application or program that you mentioned in the previous question.
    - How could you incorporate some of Bainbridge's suggestions into it?
    - What effects do you think it would have?
    - Could the suggestion itself have a negative impact?
- Then choose a representative to share your application questions to the class.

	Instructions and Advice (A)	Mitigating Error (B)	Software Displays (C)	Relieving Workload (D)
Individual: Summarize Bainbridge's points and suggestions in your own words.	Bainbridge argues that using humans merely to carry out computer instructions wastes their capacity for judgment, since those actions could be automated directly. Computer advice can explain causes, priorities, possible actions, and how to perform them, but following it may make operators slower and less coordinated than planning actions themselves. It also gives them little practice in independent problem-solving, while presenting procedures clearly remains a design challenge.	Bainbridge argues that machines should help prevent human error by checking the effects of actions rather than enforcing a particular method, unless a specific sequence is necessary. Systems should provide prompt, clear feedback so operators can recognize and correct mistakes, especially when configuring or monitoring automation. Displays can also help operators resume tasks after interruptions, while monitoring attention, such as eye movements, could help detect overlooked instruments or tunnel vision.	The author argues that software-generated displays can support operators by tailoring information to their tasks, but they can also hide abnormal conditions or cause confusion if they change unexpectedly. An operator's information needs depend on the situation, not simply their experience level, and choosing or switching displays can increase workload under time pressure. She also raises the possibility that displays which make information effortless to understand may reduce opportunities to develop deeper knowledge needed during emergencies. She therefore recommends keeping essential information continuously available, showing information needed for a single decision together, and ensuring consistency across displays. Her main point is that interfaces should support both immediate performance and the understanding needed to handle unexpected situations.	Bainbridge argues that automation can reduce human workload, but its effectiveness depends on both the amount and type of work involved. Assistance may improve performance in complex situations by freeing operators to plan or handle emergencies, yet it may offer less benefit during simpler tasks and can reduce active involvement in planning. Adapting assistance to changing workload also requires timely assessment of what the operator is handling. Crucially, operators must know which tasks the computer has taken over and how it is performing them, so responsibility remains clear. Her main point is that effective assistance requires careful coordination and attention to human trust and behaviour, rather than simply automating more decisions.

Think of your application or program that you mentioned in the previous question. How could you incorporate some of Bainbridge's suggestions into it? What effects do you think it would have? Could the suggestion itself have a negative impact?	In an AI agent system, Bainbridge's ideas could be incorporated by having the agent explain likely causes of a problem, highlight priorities, and present alternative actions with their consequences, rather than simply telling users what to do. For example, a scheduling agent could explain a calendar conflict and offer several ways to resolve it, allowing the user to choose based on their priorities. This could support informed decisions and help users practise their own judgment. However, reviewing explanations and alternatives could increase workload and slow down routine tasks, while incorrect advice could still mislead users. The system should therefore provide enough information to support meaningful choices without requiring unnecessary supervision.	In an AI agent system, Bainbridge's suggestions could be incorporated by checking the effects of users' actions and providing immediate feedback. For example, an AI scheduling assistant could flag that a meeting overlaps with another commitment without requiring users to follow a fixed scheduling process. It could also show which changes have been completed and what remains unfinished, helping users resume after interruptions. These features could help users recognize and correct mistakes while preserving flexibility. However, frequent warnings could become distracting, and incorrect warnings could reduce trust. Users might also become too dependent on these checks and overlook errors the AI fails to detect.	In an AI agent system, Bainbridge's suggestions could be incorporated by keeping the agent's current task, important warnings, and completed actions visible, while showing related information together so users do not have to switch between screens to make a decision. Users could also choose how much detail to see within a consistent layout. These features could help users understand the agent's actions, notice problems, and intervene effectively. However, displaying too much information could overwhelm users, while customization could allow them to hide details they later need. The interface should therefore offer flexibility while keeping essential status information and warnings visible.	In a restaurant kitchen management system, Bainbridge's suggestions could be incorporated by having the computer handle routine order sequencing during busy periods, allowing staff to focus on cooking and resolving problems. The display should clearly show which orders the system has prioritized and which decisions still require staff attention. This could reduce mental workload and improve coordination when many orders arrive at once. However, the system might misjudge workload or overlook practical issues, such as a cooking station being temporarily unavailable. Unexpected changes in order priority could also confuse staff, so automatic assistance should be predictable and allow easy human intervention.

3. Human-computer collaboration

By taking away the easy parts of their task, automation can make the difficult parts of the human operator's task more difficult. Several writers (Wiener and Curry, 1980; Rouse. 1981) point out that the 'Fitts list' approach to automation. assigning to man and machine the tasks they are best at, is no longer sufficient. It does not consider the integration of person and computer, nor how to maintain the effectiveness of the human operator by supporting their skills and motivation. There will always be a substantial human involvement with automated systems because criteria other than efficiency are involved, e.g. when the cost of automating some modes of operation is not justified by the value of the product, or because the public will not accept high-risk systems with no human component. This suggests that methods of human-computer collaboration need to be more fully developed. Dellner (1981) lists the possible levels of human intervention in automated decision making. This paper will discuss the possibilities for computer intervention in human decision making. These include instructing or advising the operator, mitigating their errors, providing sophisticated displays, and assisting them when task loads are high. Rouse (1981) calls these 'covert' human-computer interaction.

3.1. Instructions and advice.

Using the computer to give instructions is inappropriate if the operator is simply acting as a transducer, as the computer could equally well activate a more reliable one. Thompson (1981) lists four types of advice, about : underlying causes, relative importance, alternative actions available, and how to implement actions. When following advice the operator's reactions will be slower, and less integrated than if they can generate the sequence of activity themselves, and they are getting no practice in being 'intelligent'. There are also problems with the efficient display of procedural information.

3.2. Mitigating human error.

Machine possibilities for counteracting human error range from simple hardware interlocks to complex on-line computation. Except where specific sequences of operations must be followed it is more appropriate to place such 'checks' on the effects of actions, as this does not make assumptions about the strategy used to reach this effect. Under manual control, human operators often obtain enough feedback about the results of their actions within a few seconds to correct their own errors (Ruffell-Smith. 1979), but Wiener and Curry (1980) give examples of humans making the same types of errors in setting up and monitoring automatic equipment, when they do not get adequate feedback. This should perhaps be designed in. Kreifeldt and McCarthy (1981) give advice about displays to help operators who have been interrupted in mid-sequence. Rouse (1981) suggests computer monitoring of human eye movements to check that instrument scanning is appropriate, for example to prevent tunnel vision.

3.3. Software generated displays.

The increasing availability of soft displays on VDUs raises fascinating possibilities for designing displays compatible with the specific knowledge and cognitive processes being used in a task. This has led to such rich veins of creative speculation that it seems rather mean to point out that there are difficulties in practice.

One possibility is to display only data relevant to a particular mode of operation, such as start up routine operations, or maintenance. Care is needed however, as it is possible for an interface which is ideal for normal conditions to camouflage the development of abnormal ones (Edwards. 1981).

Goodstein (1981) has discussed process displays which are compatible with different types of operator skill, using a classification of three levels of behaviour suggested by Rasmussen (1979), i.e. skill based, rule based and knowledge based. The use of different types of skill is partly a function of the operator's experience, though the types probably do not fall on a simple continuum. Chafin (198l) has discussed how interface design recommendations depend on whether the operator is naive, novice/competent, or expert. However, he was concerned with human access to computer data bases when not under time pressure. Human-machine interaction under time pressure raises special problems. The change between knowledge-based thinking and reflex reaction is not solely a function of practice, but also depends on the uncertainty of the environment, so that the same task elements may be done using different types of skill at different times. It could therefore confuse rather than help the operator to give them a display which is solely a function of their overall skill level. Non-time-stressed operators, if they find they have the wrong type of display, might themselves request a different level of information. This would add to the work load of someone making decisions which are paced by a dynamic system. Rouse (1981) has therefore suggested that the computer might identify which type of skill the operator is using, and change the displays (he does not say how this might be done). We do not know how confused operators would be by display changes which were not under their own control. Ephraph and Young (1981) have commented that it takes time for an operator to shift between activity modes, e.g. from monitoring to controlling, even when these are under the person's control, and one assumes that the same problems would arise with changes in display mode. Certainly a great deal of care would be needed to make sure that the different displays were compatible. Rasmussen and Lind's recent paper (1981) was about the different levels of abstraction at which the operator might be thinking about the process, which would define the knowledge base to be displayed. Again, although operators evidently do think at different levels of complexity and abstraction at different times, it is not clear that they would be able to use, or choose, many different displays under time stress.

Some points were made above about the problems of operators who have learned to work with computer generated displays, when these displays are no longer available in abnormal conditions. Recent research on human memory (Craik, 1979) suggests that the more processing for meaning that some data has received the more effectively it is remembered. This makes one wonder how much the operator will learn about the structure of the process if information about it is presented so successfully that they does not have to think about it to take it in. It certainly would be ironic if we find that the most compatible display is not the best display to give to the operator after all! (As usual with display choice decisions this would depend on the task to be done. A highly compatible display always supports rapid reactions. These points speculate whether they also support acquisition of the knowledge and thinking skills needed in abnormal conditions.)

A few practical points can be suggested. There should be at least one source of information permanently available for each type of information which cannot be mapped simply onto others, e.g. about layout of plant in space as opposed to its functional topology. Operators should not have to page between displays to obtain information about abnormal states in parts of the process other than the one they are currently thinking about, nor between displays giving information needed within one decision process. Research on sophisticated displays should concentrate on the problems of ensuring compatibility between them, rather than finding which independent display is best for one particular function without considering its relation to information for other functions. To end on a more optimistic note, software displays offer some interesting possibilities for enriching the operator's task by allowing them to design their own interface.

3.4. Relieving human workload.

A computer can be used to reduce human workload either by simplifying the operator's decisions, as above, or by taking over some of the decision making. The studies which have been done on this show that it is a complex issue. Ephrath and Young (1981) found that overall control performance was better with manual control of a single loop, but was also better with an autopilot in the complex environment of a cockpit simulator. This suggests that aiding is best used at higher work loads. However, the effect of the type of aiding depends on the type of workload. Johannsen and Rouse (1981) found that pilots reported less depth of planning under autopilot in abnormal environmental conditions, presumably because the autopilot was dealing with the conditions, but more planning under emergency aircraft conditions, where they suggest that the autopilot frees the pilot from on-line control so he can think about other things. Chu and Rouse (19791 studied a situation with both computer aiding and autopilot. They arranged for the computer to take over decision making when the pilot had a queue of one other task item to be dealt with and the pilot was controlling manually, or after a queue of three items if the autopilot was controlling. The study by Enstrom and Rouse (1977) makes it clear why Rouse (l98l) comments that more sophisticated on-line methods of adapting computer aiding to human workload will only be possible if the workload computations can be done in real time. It would be rash to claim it as an irony that the aim of aiding human limited capacity has pushed computing to the limit of its capacity, as technology has a way of catching up with such remarks.

Enstrom and Rouse also make the important point that the human being must know which tasks the computer is dealing with and how. Otherwise the same problems arise as in human teams in which there is no clear allocation of responsibility. Sinaiko (1972) makes a comment which emphasizes the importance of the human operator's perception of the computer's abilities: "when loads were light, the man appeared willing to let the computer carry most of the assignment responsibility: when loads were heavy, the man much more often stepped in and over-rode the computer". Evidently, quite apart from technical considerations, the design of computer aiding is a multidimensional problem.